

GROUP NAME	GROUP MEMBERS (Name, Surname, email)	CFU (6/9)
Group03	Rachele Gorghetto, rachele.gorghetto@studenti.unipd.it Lisa Rallo, lisa.rallo.1@studenti.unipd.it Leonardo Rizzeri, leonardorizzeri@gmail.com Camilla Di Gioacchino, camilla.digioacchino@studenti.unipd.it	9

Imaging for Neuroscience Homework report

ANATOMICAL PROCESSING OF THE T1W IMAGE

To extract reliable structural priors for multimodal integration, the brain-extracted T1-weighted (T1w) anatomical image was automatically segmented, and bias field corrected, into three distinct tissue classes using the **FSL FAST**^[1]. This automated tissue segmentation algorithm models spatial intensity distributions via a hidden Markov random field model to classify voxels into Cerebrospinal Fluid (CSF), Grey Matter (GM), and White Matter (WM) partial volume estimation (PVE) maps.

To ensure optimal tissue classification and remove voxels with high tissue ambiguity, specific probability thresholds were applied to the continuous PVE maps. Based on empirical visual inspection and standard protocols, the threshold parameters were defined as follows:

- a strict threshold of **0.6** was applied to the CSF PVE map;
- a threshold of **0.4** was applied to both the GM and WM PVE maps.

Following this thresholding step, the segmented tissue profiles were binarized to generate finalized, anatomically precise binary masks suitable for downstream spatial coregistration and region-of-interest (ROI) masking. The resulting spatial distributions are overlaid on the structural template as detailed in **Figure 1**.

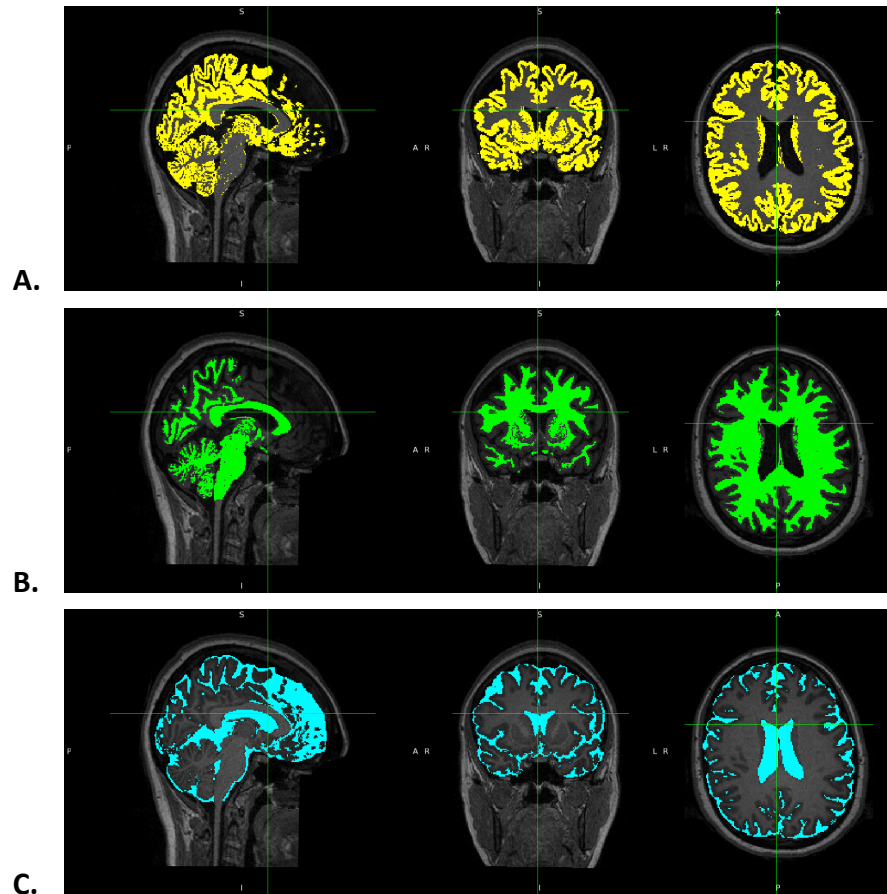


Figure 1. Brain tissue segmentation masks obtained from the brain-extracted T1-weighted image. Orthographic views (sagittal, coronal, and axial) showing the binarized segmentation masks. The panels illustrate the three segmented tissue classes: **(A)** Grey Matter masked in yellow, highlighting the ventricular system and subarachnoid space; **(B)** White Matter masked in green, encompassing the cerebral cortex and deep subcortical nuclei; **(C)** Cerebrospinal Fluid masked in cyan, capturing the underlying myelinated fiber tracts.

PROCESSING OF THE [18F]FDG-PET DATA

The dynamic [18F]FDG-PET acquisition consisted of a 4D dataset encompassing 31 time frames over a total acquisition time of 45 minutes, with an anisotropic voxel size of $1.0431 \times 1.0431 \times 1.0156 \text{ mm}^3$. Cortical regions were delineated by registering the Schaefer 100-parcel 7-network atlas combined with 18 cerebellar ROIs from the AAL3 atlas to the PET reference space. To extract the Time-Activity Curves (TACs), the atlas was masked with the registered GM mask to restrict metabolic quantification specifically to grey matter tissue. The temporal dynamics of the PET tracer concentration across the 7 major functional macro-networks and the reference region are plotted in **Figure 2**.

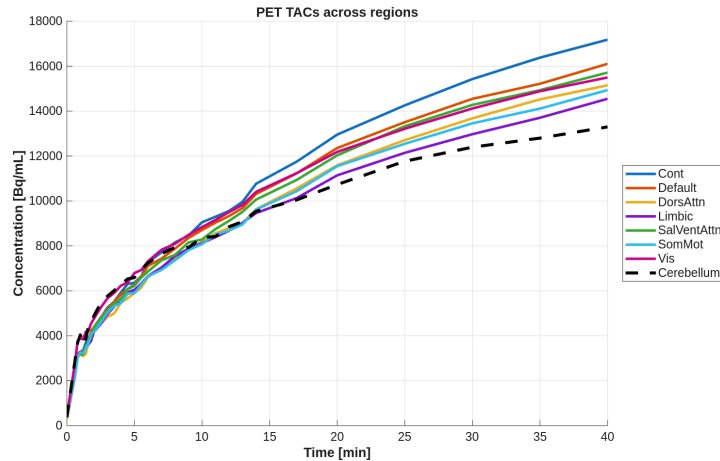


Figure 2. Comparison of time-activity curves (TACs) between the reference region and cortical ROIs. Temporal dynamics of the PET signal across the different extracted cerebral macro-networks. The black dashed line highlights the cerebellum (Cerebellum), utilized as the reference region for kinetic modeling in comparison to the target areas (colored lines).

A late-phase time window of 25–35 minutes was selected to compute the Standardized Uptake Value ratio (SUVr). From a kinetic modeling perspective, the choice of this specific interval requires rigorous biochemical justification due to the intrinsic properties of [18F]FDG. Unlike reversible receptor-binding radiotracers, [18F]FDG (fluorodeoxyglucose) acts as a glucose analog subject to "metabolic trapping": once [18F]FDG is delivered into the brain, it is trapped in the tissue in its phosphorylated form (the slow dephosphorylation can be considered negligible), following the deoxyglucose path^[2]. Consequently, the target tissue TACs exhibit a monotonic, non-plateauing upward trajectory, while the vascular pool undergoes continuous clearance as the tracer is sequentially sequestered by metabolically active regions. Selecting a late-phase window towards the end of the 40-minute acquisition is justified by three interconnected factors:

1. Vascular Clearance and SUVr Stabilization: By 25–35 minutes post-injection, the initial rapid vascular clearance phase has reached completion, and the target tissue accumulation enters a linear phase.
2. Independence from Perfusion/Blood Flow: Utilizing a later window ensures that the non-specifically bound, free tracer has been sufficiently washed out, meaning the recorded signal is highly proportional to net metabolic trapping.
3. Methodological Alignment with Patlak Kinetics: In tracer kinetic literature, the formal justification relies on the Patlak graphical method for irreversible systems, which establishes that the influx slope becomes constant after an initial transient equilibrium period.

The target regional activity in this stabilized late-phase window was divided by the average concentration extracted from the whole cerebellum (utilized as the reference tissue), yielding the final cortical SUVr values depicted via the stem plot in **Figure 3**.

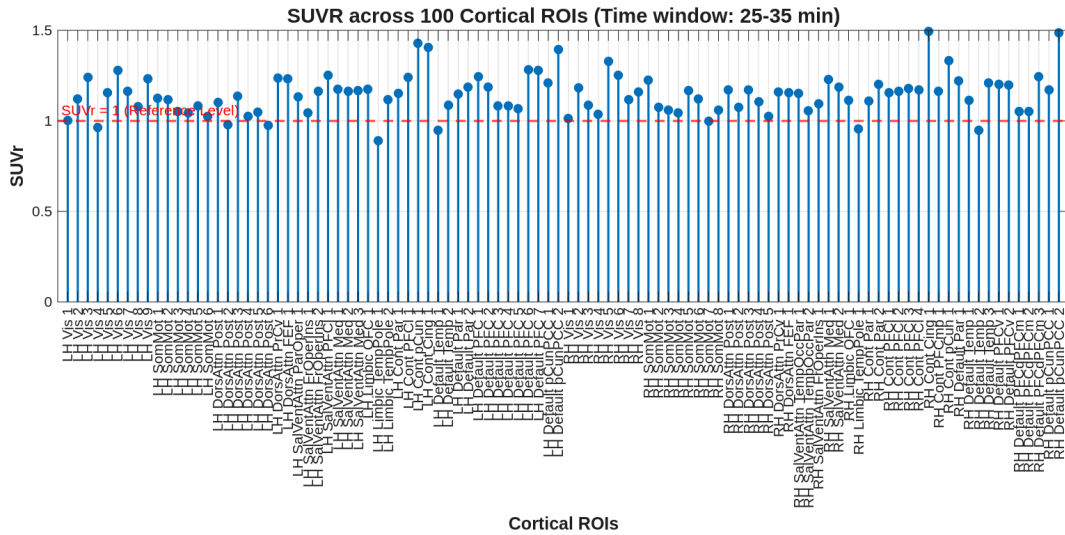


Figure 3. Distribution of SUVR values across 100 cortical ROIs (Time window: 25-35 min). Stem plot representation of the standardized uptake value ratio (SUVR) for each extracted cortical region of interest (reference region excluded). The red dashed line denotes the baseline reference level (SUVR = 1), highlighting the relative tracer accumulation in target areas compared to the cerebellum.

PROCESSING OF RSfMRI DATA

The resting-state functional MRI (rs-fMRI) sequence comprised 300 volumes acquired on a 3T scanner (TR = 2.0 s, voxel size = $3.0 \times 3.0 \times 3.6 \text{ mm}^3$). Initial preprocessing included slice-timing correction and rigid-body motion correction, with the mean functional volume chosen as the reference frame. To thoroughly assess data quality, patient head motion parameters were investigated across the entire dynamic run. The estimated rigid-body translations and rotations were plotted to evaluate structural displacement (**Figure 4A**), while the Framewise Displacement (FD) was vectorially computed according to the Power et al. (2012)^[3] formulation to track frame-by-frame instantaneous motion (**Figure 4B**). The dataset exhibited excellent low-motion characteristics with 0% of the volumes exceeding the critical quality control threshold of 0.5 mm, confirming that no volume scrubbing or exclusion was required.

Multimodal alignment was achieved by inverting the pre-computed fMRI-to-T1w linear registration matrix and resampling the structural CSF, GM, and WM tissue masks into native fMRI space. To systematically eliminate non-neural physiological confounds and motion artifacts from the BOLD signals, a voxelwise nuisance regression was implemented via a Generalized Linear Model (GLM). A fundamental step in this pipeline involved restricting the analysis specifically to intracranial voxels to prevent background noise amplification. An empirical intensity threshold of 370 was selected to define the operational brain mask. This thresholding approach is highly critical in functional pipelines to robustly separate low-intensity non-brain voxels and background scanner noise from the true intraparenchymal BOLD signal, ensuring that peripheral artifacts do not distort downstream voxelwise calculations^[4]). The GLM design matrix was subsequently constructed using 22 distinct regressors (**Figure 5**).

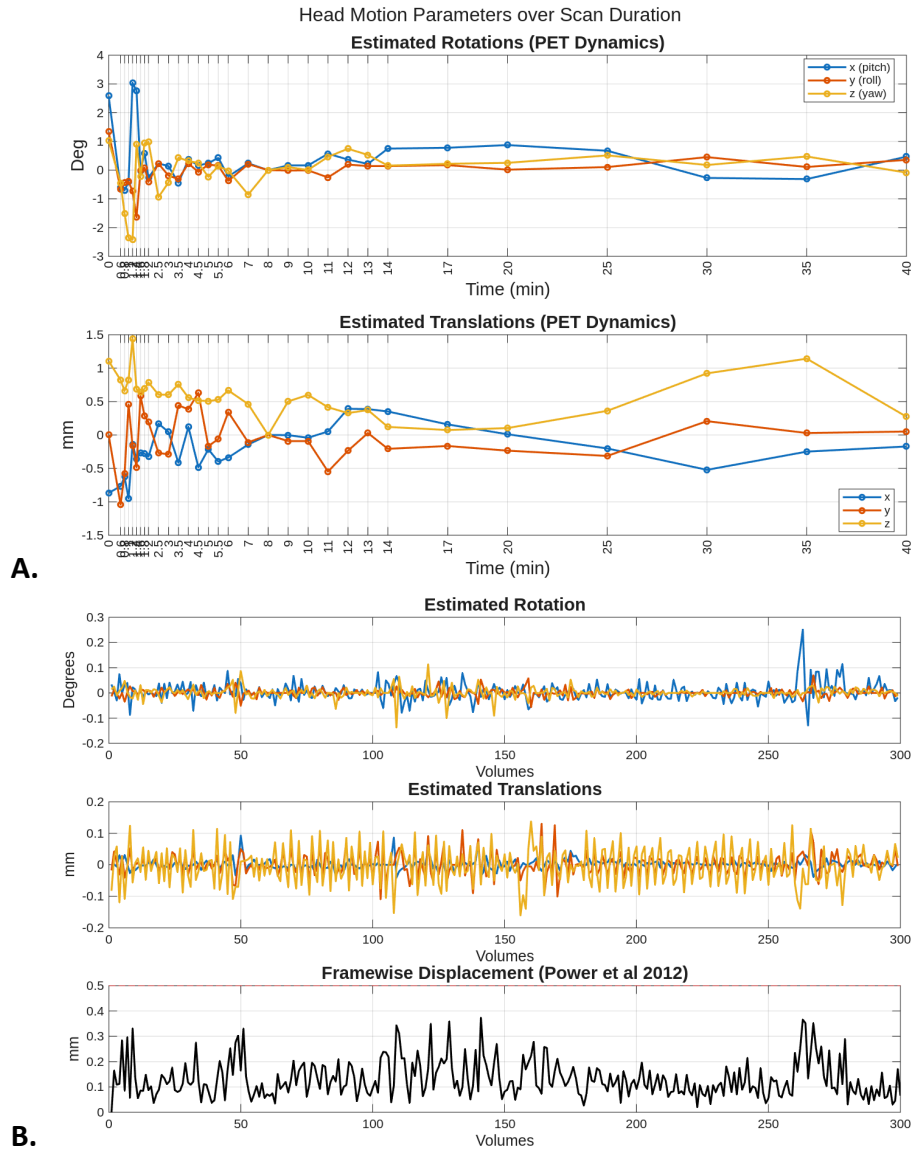


Figure 4. (A) Dynamic head motion parameters over the PET scan timeline. Estimated rotations (top) and translations (bottom) plotted directly against the PET acquisition time frames in minutes. The uneven sampling density along the x-axis reflects the shorter frame durations at the start of the scan and the longer, consolidated time frames in the later phases of the dynamic acquisition. The red dashed line denotes the baseline reference level ($SUV_r = 1$), highlighting the relative tracer accumulation in target areas compared to the cerebellum. **(B) Volume-by-volume head motion and Framewise Displacement (FD).** The top two subplots display the estimated rigid-body motion parameters for each acquired volume, showing rotations in degrees and translations (x , y , z) in millimeters. The bottom subplot illustrates the corresponding Framewise Displacement calculated using the Power et al. (2012) formulation.

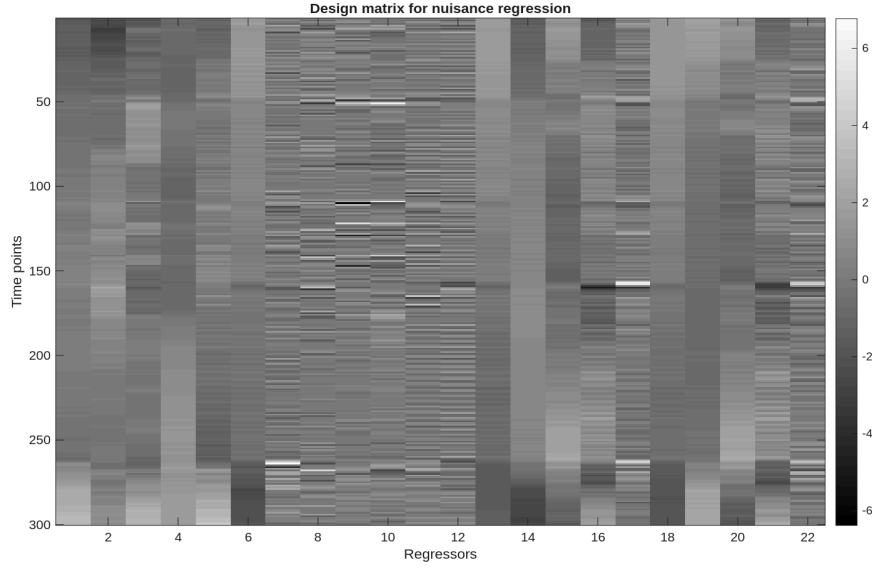


Figure 5. Design matrix for nuisance regression. Graphical representation of the design matrix utilized to model and regress out confounding factors from the functional data. The matrix includes 300 time points (rows) and 22 distinct nuisance regressors (columns). The grayscale intensity bar on the right indicates the normalized weight/value of each regressor across the scan duration.

Following nuisance regression, the residuals underwent temporal bandpass filtering using a 3rd-order Butterworth high-pass filter with a cut-off frequency of 1/128 Hz to eliminate slow scanner drifts while preserving spontaneous low-frequency BOLD fluctuations (**Figure 6**).

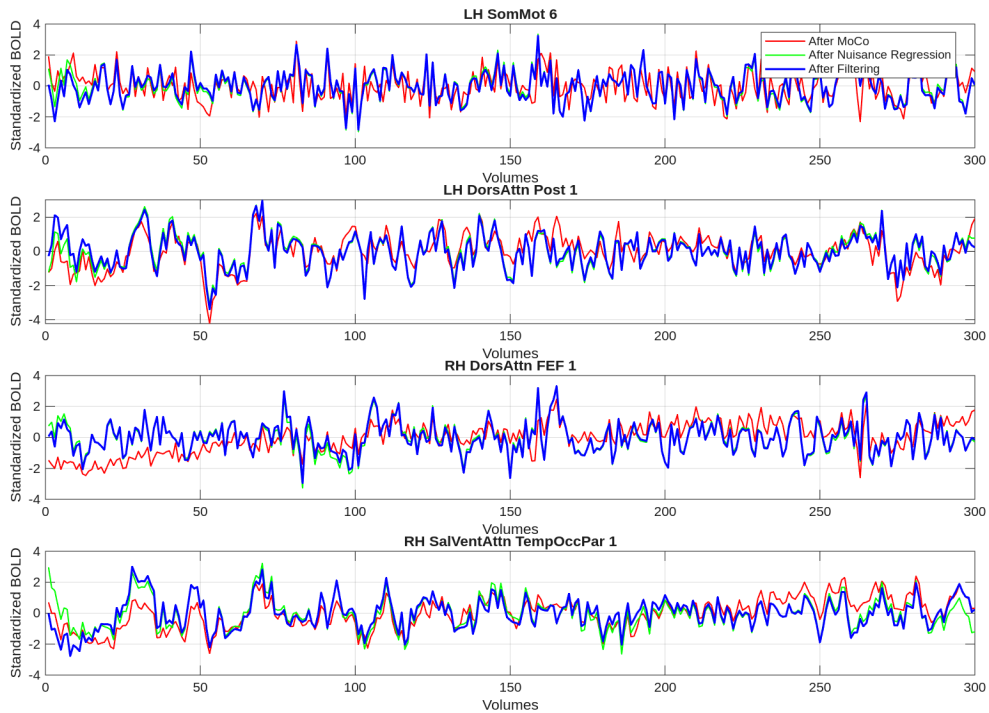


Figure 6. BOLD signal time series across preprocessing stages for selected cortical ROIs. Comparison of the standardized BOLD signal extracted from four target regions (LH_SomMot_6, LH_DorsAttn_Post_1, RH_DorsAttn_FEF_1, RH_SalVentAttn_TempOccPar_1) across 300 volumes. The curves track the signal evolution through consecutive preprocessing steps: after motion correction (red), following nuisance regression (green), and after the application of a 3rd-order Butterworth high-pass filter with a 1/128 Hz cut-off frequency (blue).

Static functional connectivity (FC) was quantified by computing pairwise Pearson correlation coefficients (r) across the 100 cortical ROIs of the Schaefer atlas. The resulting coefficients were stabilized via Fisher z -transformation and statistically thresholded using a False Discovery Rate (FDR) correction ($\alpha = 0.05$) to isolate significant functional connections (**Figure 7**), from which nodal centrality metrics (Node Degree and Node Strength) were computed (**Figure 8**).

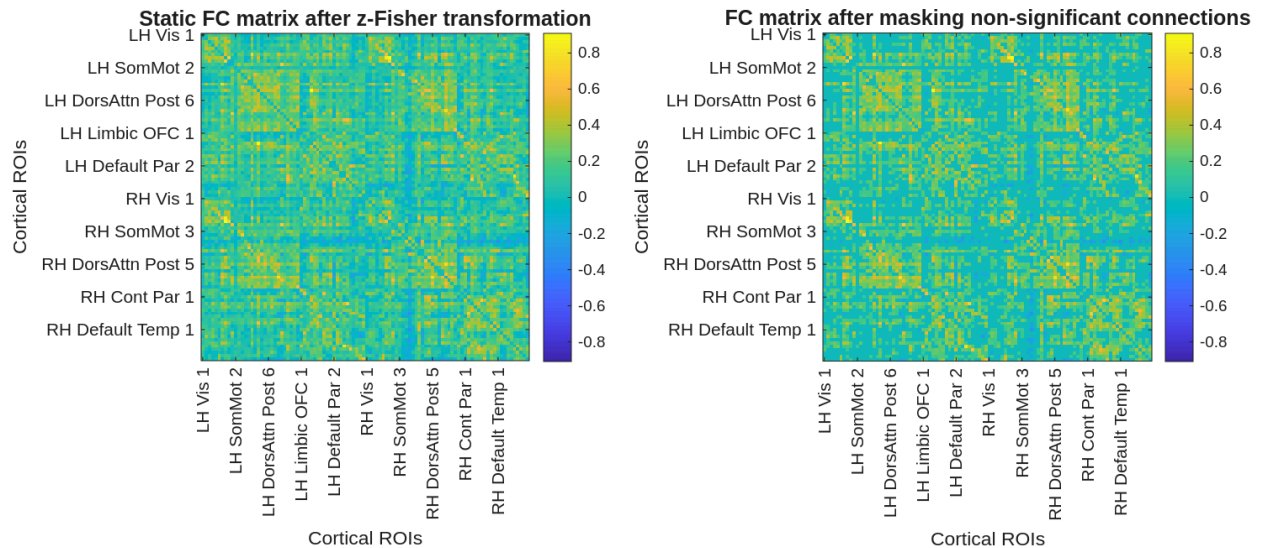


Figure 7. Static functional connectivity matrices before and after statistical thresholding. (Left) Whole-brain static functional connectivity (FC) matrix computed via Pearson correlation across cortical regions of interest and stabilized using the Fisher z -transformation. **(Right)** Corresponding FC matrix after masking out non-significant connections. The color bars represent the strength and direction of the functional coupling (z -values), ranging from positive correlation (yellow) to negative correlation (blue), with non-significant coefficients zeroed out in the masked matrix.

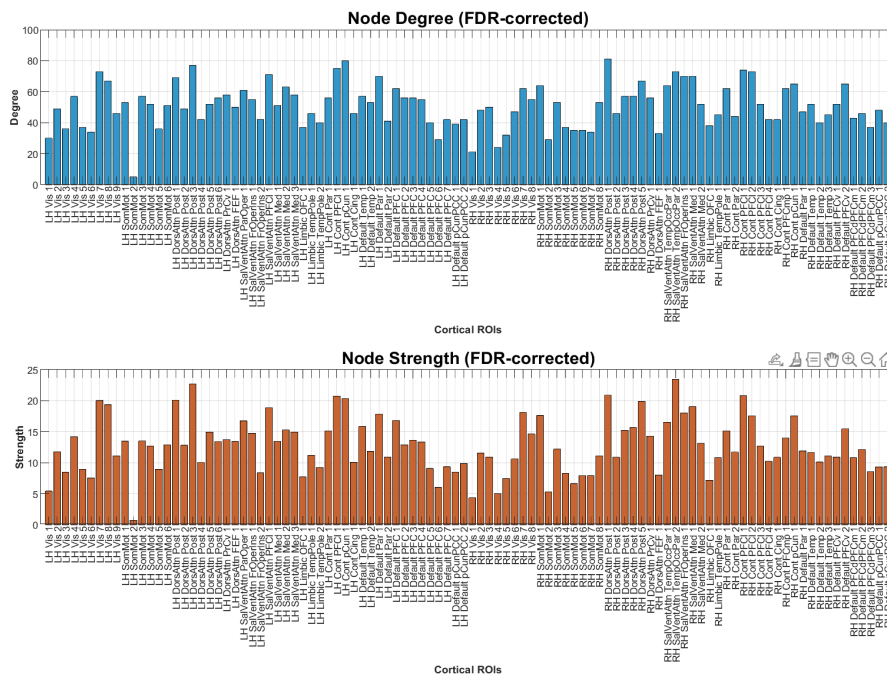


Figure 8. Node degree and node strength distribution across cortical ROIs. Bar plot representation of network centrality metrics derived from the FDR-corrected static functional connectivity matrix across the 100 cortical regions of interest. The top panel illustrates the Node Degree, representing the total number of significant functional connections sustained by each node. The bottom panel displays the Node Strength, representing the sum of the absolute weights of those significant connections. Prominent peaks highlight hub regions with the highest degree and strength, indicating central structural-functional integration points within major cortical networks (e.g., LH_DorsAttn_Post_2, LH_Cont_PFC_1, and RH_SalVentAttn_TempOccPar_2).

To characterize the low-dimensional macro-scale topographic organization of the functional connectome, a non-linear manifold learning analysis was performed on the full FC matrix using the BrainSpace toolbox. Applying Laplacian Eigenmaps embedding with a Gaussian kernel, the high-dimensional connectivity profiles were projected into a continuous functional gradient space (**Figure 9**). The distribution of cortical regions within this low-dimensional embedding perfectly mirrors the foundational framework established by *Margulies et al. (2016)*^[5]. Specifically, Gradient 1 delineates a continuous global axis that systematically segregates primary, unimodal sensory-motor regions (Visual and Somatomotor networks) from transmodal association areas anchored by the Default Mode Network (DMN), capturing the hierarchical macro-scale transition from extrinsic stimulus-driven processing to abstract, internally-directed cognition.

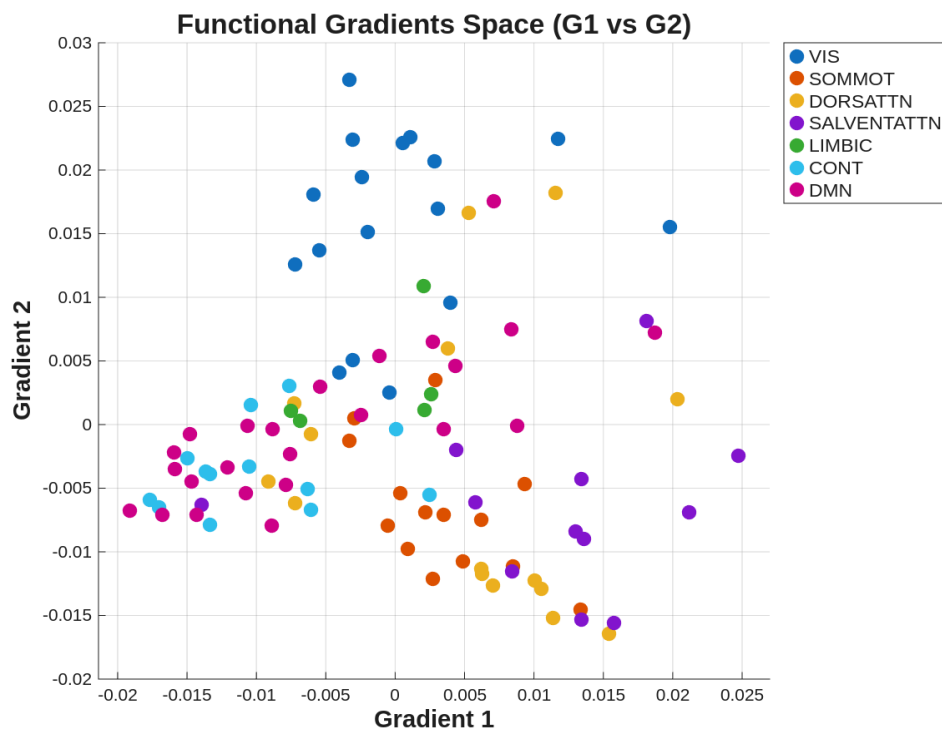


Figure 9. Low-dimensional macro-scale functional gradient space. Scatterplot of the first two principal components (Gradient 1 vs. Gradient 2). Each dot represents a cortical region of interest (ROI), color-coded according to its macro-scale functional network assignment based on the Schaefer 100-parcel 7-network atlas.

MULTIMODAL INTEGRATION

For the PET-rsfMRI integration, regional FDG-PET SUVR values were compared with the rs-fMRI-derived degree, strength, and first functional gradient, as required by the assignment. Since the analysis was performed at the ROI level, each point in **Figure 10** represents one cortical region. Spearman correlation was used to quantify these associations because degree is a discrete graph metric and because the relationship between glucose metabolism and functional organization was not assumed to be strictly linear or normally distributed. This makes Spearman correlation more appropriate than a purely linear Pearson correlation for this exploratory analysis.

The results showed no relevant association between SUVR and degree ($\rho = 0.03$, $p = 0.732$) or between SUVR and strength ($\rho = 0.01$, $p = 0.902$). Therefore, in this subject, cortical regions with more significant functional connections did not show higher glucose metabolism. In contrast, SUVR showed a moderate and significant negative association with Gradient 1 ($\rho = -0.39$, $p < 0.001$). This suggests that glucose metabolism was not mainly related to graph-based hubness, but rather varied along the principal axis of cortical functional organization.

This pattern is plausible considering previous literature. Tomasi et al. (2013)^[6] showed that metabolic hubs can be identified using [18F]FDG-PET and that brain metabolism is related to network organization. However, Volpi et al (2024)^[7] emphasized that the relationship between glucose metabolism and resting-state brain activity is complex, and that rs-fMRI measures cannot fully explain the regional variability of FDG uptake. In this context, the absence of correlation with degree and strength may reflect the fact that SUVR and BOLD functional connectivity describe related but non-equivalent physiological properties. Overall, our findings do not support a direct relationship between functional hubness and metabolic demand in this single subject, but they suggest that glucose metabolism may be organized along broader cortical functional gradients. However we should note that we are doing a study over a single subject with no information on his health, so we must consider that every conclusion we may try to provide will lack statistical significance and could be biased by some pathological condition of the subject .

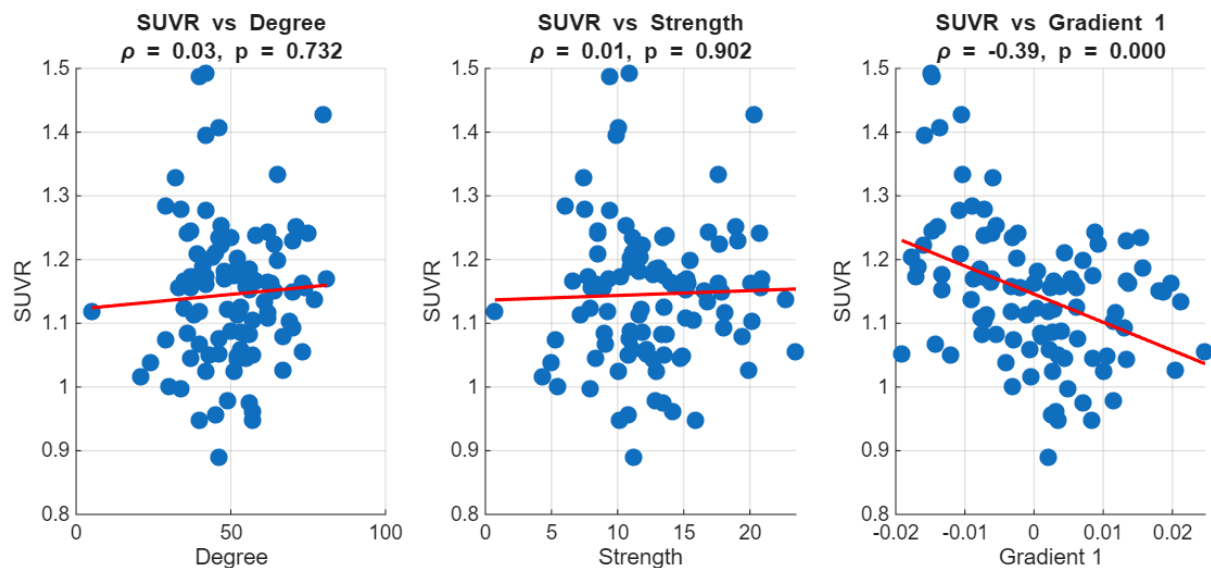


Figure 10. Relationship between regional glucose metabolism and rs-fMRI-derived functional organization across cortical ROIs. Scatter plots show FDG-PET SUVR as a function of degree, strength, and the first functional gradient, from left to right. Spearman's correlation coefficient (ρ) and the corresponding p -value are reported for each comparison. No significant association was observed between SUVR and degree or strength, whereas SUVR showed a significant association with Gradient 1.

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